

```
SQL> create table sale(sid integer primary key,sname
varchar(30),city varchar(30),comission integer);
```

Table created.

```
SQL insert into sale values('1','ibrahim','bangalore','5000');
```

1 row created.

```
SQL> insert into sale values('2','husain','hubli','4000');
```

1 row created.

```
SQL> insert into sale values('3','mazhar','haveri','3500');
```

1 row created.

```
SQL> insert into sale values('4','aftab','mumbai','3000');
```

1 row created.

```
SQL> insert into sale values('5','zameer','pune','2500');
```

1 row created.

```
SQL> select * from sale;
```

SID	SNAME	CITY	COMISSION
1	ibrahim	bangalore	5000
2	husain	hubli	4000
3	mazhar	haveri	3500
4	aftab	mumbai	3000
5	zameer	pune	2500

```
SQL> create table cust(cid integer primary key,cname
varchar(30),ccity varchar(20),grade integer,sid
integer,foreign key(sid) references sale(sid));
```

Table created.

```
SQL> insert into cust values('1','vinod','hubli','3','1');
```

1 row created.

```
SQL> insert into cust values('2','aftab','bangalore','2','1');
```

1 row created.

```
SQL> insert into cust values('3','sadiq','mumbai','5','2');
```

1 row created.

```
SQL> insert into cust values('4','altaf','bangalore','3','3');
```

1 row created.

```
SQL> insert into cust values('5','afnan','pune','1','3');
```

1 row created.

```
SQL> select * from cust;
```

CID	CNAME	CCITY	GRADE	SID
1	vinod	hubli	3	1
2	aftab	bangalore	2	1
3	sadiq	mumbai	5	2
4	altaf	bangalore	3	3
5	afnan	pune	1	3

```
SQL> create table ord(ordno integer primary key,pamt
integer,odate date,sid integer,foreign key(sid)references
sale(sid),cid integer,foreign key(cid) references cust(cid));
```

Table created.

```
SQL>insert into ord values('1','25000','10-dec-2019','1','1');
```

1 row created.

```
SQL> insert into ord values('2','2000','19-jan-2021','1','1');
```

1 row created.

```
SQL>insert into ord values('3','35000','30-june-2023','2','2');
```

1 row created.

```
SQL> insert into ord values('4','8500','28-oct-2021','3','2');
```

1 row created.

```
SQL> insert into ord values('5','10000','12-may-2023','4','5');
```

1 row created.

```
select * from orde;
```

ORDNO	PAMT	ODATE	SID	CID
1	25000	10-DEC-19	1	1
2	2000	19-JAN-21	1	1
3	35000	30-JUN-23	2	2
4	8500	28-OCT-21	3	2
5	10000	12-MAY-23	4	4

i. Count the customers with grades above Bangalore's average.

```
SQL> select count(cid) from cust where grade>(select
avg(grade) from cust where ccity='bangalore');
```

```
COUNT(CID)
-----
3
```

ii. Find the name and numbers of all salesmen who had more than one customer.

```
SQL> select s.sname,s.sid from sale s,cust c where s.sid=c.sid
group by s.sname,s.sid having count(c.cid)>1;
```

SNAME	SID
ibrahim	1
mazhar	3

iii. List all salesmen and indicate those who have and don't have customers in their cities (Use UNION operation.)

```
(select s.sid,s.sname,c.cname from sale s,cust c where
s.city=c.ccity and s.sid=c.sid)union(select
s1.sid,s1.sname,'no customer'from sale s1,cust c1 where
s1.city=c1.ccity and s1.sid=c1.sid);
```

SID	SNAME	CNAME
1	ibrahim	aftab
1	ibrahim	no customer

iv. Create a view that finds the sales man who has the customer with the highest order of a day.

```
SQL> create view high_order_day as select
o.odate,s.sid,s.sname,c.cname,o.pamt from cust232 o,sale
s,cust c where o.sid=s.sid and c.cid=o.cid;
```

View created.

```
SQL> select * from high_order_day;
```

ODATE	SID	SNAME	CNAME	PAMT
10-DEC-19	1	ibrahim	vinod	25000
19-JAN-21	1	ibrahim	vinod	2000
30-JUN-23	2	husain	aftab	35000
28-OCT-21	3	mazhar	aftab	8500
12-MAY-23	4	aftab	afnan	10000

v. Demonstrate the DELETE operation by removing salesman with id 1000. All his orders must also be deleted.

```
SQL> delete from sale where sid='5';
```

1 row deleted.

```
SQL> select * from sale;
```

SID	SNAME	CITY	COMISSION
1	ibrahim	bangalore	5000
2	husain	hubli	4000
3	mazhar	haveri	3500
4	aftab	mumbai	3000

Write PL/SQL program to demonstrate %ROWCOUNT attribute.

```
SQL> declare
2     total_rows number(2);
3     begin
4     update sale set comission=comission+500;
5     if sql%notfound then
6     dbms_output.put_line('no customers selected');
7     elsif sql%found then
8     total_rows:=sql%rowcount;
9     dbms_output.put_line(total_rows || 'salesman updated');
10    end if;
11    commit;
12    end;
13    /
```

5salesman updated

PL/SQL procedure successfully completed.

```
SQL> select * from sale;
```

SID	SNAME	CITY	COMISSION
1	ibrahim	bangalore	5500
2	husain	hubli	4500
3	mazhar	haveri	4000
4	aftab	mumbai	3500
5	zameer	pune	3000